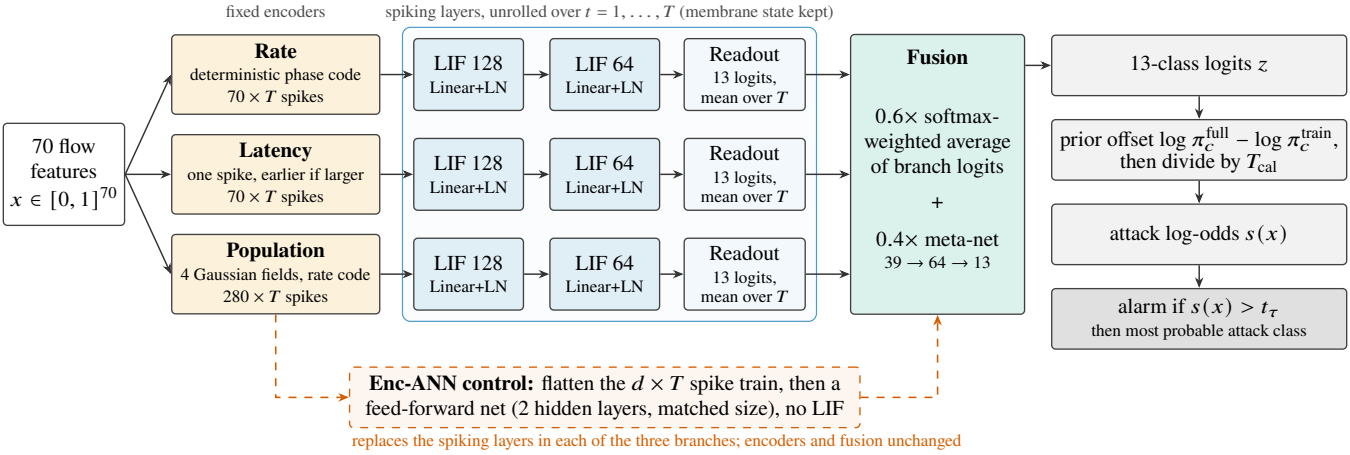
fixed encoders
spiking layers, unrolled over $t = 1, \ldots, T$ (membrane state kept)
70 flow features $x \in [0, 1]^{70}$
Rate
deterministic phase code
$70 \times T$ spikes
Latency
one spike, earlier if larger
$70 \times T$ spikes
Population
4 Gaussian fields, rate code
$280 \times T$ spikes
LIF 128
Linear+LN
LIF 64
Linear+LN
Readout
13 logits, mean over T
Fusion
0.6× softmax-weighted average of branch logits
+
0.4× meta-net
$39 \to 64 \to 13$
13-class logits z
prior offset $\log \pi_c^{\text{full}} - \log \pi_c^{\text{train}}$, then divide by T_{cal}
attack log-odds $s(x)$
alarm if $s(x) > t_\tau$
then most probable attack class
Enc-ANN control: flatten the $d \times T$ spike train, then a feed-forward net (2 hidden layers, matched size), no LIF
replaces the spiking layers in each of the three branches; encoders and fusion unchanged